CRYPTOGRAPHY AND NETWORK SECURITY LAB

(Course Code: 23UPCSC1E14)

A laboratory record submitted to Periyar University, Salem

In partial fulfillment of the requirements for the degree of

MASTER OF COMPUTER APPLICATION

By

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PERIYAR UNIVERSITY

(NACC `A++` Grade with CGPA 3.61) – NIRF RANK 56

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SALEM – 636 011.

(APRIL - 2024)

CERTIFICATE

This is to certify that the Programming Laboratory entitled "**CRYPTOGRAPHY AND NETWORK SECURITY LAB (23UPCSC1E14)"** is a bonafide record work done by Mr. /Ms. _____Register No: _____as partial fulfillment ofthe requirements for the degree of Master of Computer Application, in the Department of Computer Science, Periyar University, Salem, during the AcademicYear 2024-2025.

Staff In-charge Head of the Department

Submitted for the practical examination held on……..……

Internal Examiner External Examiner

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9.		Implement signature scheme – Digital Signature Standard		
10.		Identify and capture the user name and password in a same network using wireshark		

SOURCE CODE:

```
def encryptDecrypt(inpString):

    xorKey='H'

    length=len(inpString)

    for i  in range(length):

        inpString =
    (inpString[:i]+chr(ord(inpString[i])^ord(xorKey))+inpString[i+1:])

        print(inpString[i],end='')

    return inpString

if __name__=='__main__':

    sampleString=input("Enter the plain Text:")

    print("encrypted string:",end='')

    sampleString=encryptDecrypt(sampleString)

    print('\n')

    print("decrypted String:",end='')

  encryptDecrypt(sampleString)
```

OUTPUT:

```
Enter the plain Text:Vulnerability
encrypted string:=$&-:)*!$!<l

decrypted String:Vulnerability
```

SOURCE CODE

```
def encrypt(text,s):
    result=''
    for i in range(len(text)):
        char=text[i]
        if (char.isupper()):
            result++chr((ord(char)+s-65)%26+65)


        else:
            result+=chr((ord(char)+s-97)%26+97)
    return result
text=input("Enter the plain text to convert ceaser cipher:")
s=5
print("Text:",text)
```

```
print("Shift:",str(s))

print("Cipher:",encrypt(text,s))

def decrypt(text,s):

    result=""

    for i in range(len(text)):

        char=text[i]

        if(char.isupper()):

            result+=chr((ord(char)+s-65)%26+65)

        else:

            result+=chr((ord(char)+s-97)%26+97)

    return result

text=input("Enter the ceaser cipher to convert plain text:")

s=26-5

print("Text:"+text)

print("Shift:"+str(s))

print("Cipher:"+decrypt(text,s))
```

OUTPUT:

```
Enter the plain text to convert ceaser cipher:cryptography
Text: cryptography
Shift: 5
Cipher: hwduytlwfumd
Enter the ceaser cipher to convert plain text:hwduytlwfumd
Text:hwduytlwfumd
Shift:21
Cipher:cryptography
```

SOURCE CODE:

```
keyMatrix=[[0]*3 for i in range(3)]

messageVector=[[0] for i in range(3)]

cipherMatrix=[[0] for i in range(3)]



def getKeyMatrix(key):

    k=0

    for i in range(3):

        for j in range(3):

            keyMatrix[i][j]=ord(key[k])%65

            k+=1

def encrypt(messageVector):

    for i in range(3):

        for j in range(1):

            cipherMatrix[i][j]=0
```

```
        for x in range(3):
            cipherMatrix[i][j]+=(keyMatrix[i][x]*messageVector[x][j])
        cipherMatrix[i][j]=cipherMatrix[i][j]%26


def HillCipher(message,key):
    getKeyMatrix(key)
    for i in range(3):
        messageVector[i][0]=ord(message[i])%65


    encrypt(messageVector)
    CipherText=[]
    for i in range(3):
        CipherText.append(chr(cipherMatrix[i][0]+65))
    print("Ciphertext:",".join(CipherText))
```

```
def main():

    message=str(input("Enter the plain text:"))

    key=str(input("Enter the key:"))

    HillCipher(message,key)



if __name__=='__main__':

    main()
```

OUTPUT:

```
Enter the plain text:crpyto
Enter the key:blockchain
Ciphertext: JCQ
```

SOURCE CODE:

```
import os

def  generate_key(length):

    return os.urandom(length)

def vernam_encrypt(plaintext, key):

    if len(plaintext) != len(key):

        raise ValueError("The key must be the same length as the
    plaintext.")

    ciphertext = bytes([p ^ k for p, k in zip(plaintext.encode('utf-8'),
    key)])

    return ciphertext

def  vernam_decrypt(ciphertext, key):

    if len(ciphertext) != len(key):

        raise ValueError("The key must be the same length as the
    ciphertext.")

     plaintext = ".join([chr(c ^ k) for c, k in zip(ciphertext, key)])
```

```
    return plaintext

def main():

    plaintext = "This is another example of the Vernam cipher!"

    key = generate_key(len(plaintext))

    print(f"Generated Key (in bytes): {key.hex()}")  # Display key in
    hexadecimal for better readability

    ciphertext = vernam_encrypt(plaintext, key)

    print(f"Encrypted Ciphertext (in bytes): {ciphertext.hex()}")

    decrypted_text = vernam_decrypt(ciphertext, key)

    print(f"Decrypted Plaintext: {decrypted_text}")

if __name__ == "__main__":

    main()
```

OUTPUT:

```
Generated Key (in bytes): 43e6e5436afbf3eb7b15d7004bc92035b4625850fb03dca13f174b
a474d7b76a0ae6f8261f2b7e257bed730b7c
Encrypted Ciphertext (in bytes): 178e8c304a9280cb1a7bb87423ac5215d11a393d8b6fb98
150716bd01cb2973c6f949647720b1d4c0b8516795d
Decrypted Plaintext: This is another example of the Vernam cipher!
```

SOURCE CODE:

```
import string

all_letters=string.ascii_letters

dict1={}

key=4

for i in range(len(all_letters)):

    dict1[all_letters[i]]=all_letters[(i+key)%len(all_letters)]

plain_txt=str(input("Enter your plain_txt:"))

cipher_txt=[]

for char in plain_txt:

    if char in all_letters:

        temp=dict1[char]

        cipher_txt.append(temp)

    else:

        temp=char
```

```
        cipher_txt.append(temp)
cipher_txt=''.join(cipher_txt)
print("Cipher Text is :",cipher_txt)
dict2={}
for i in range(len(all_letters)):
    dict2[all_letters[i]]=all_letters[(i-key)%(len(all_letters))]
decrypt_txt=[]
for char in cipher_txt:
    if char in all_letters:
        temp=dict2[char]
        decrypt_txt.append(temp)
    else:
        temp=char
        decrypt_txt.append(temp)
decrypt_txt=''.join(decrypt_txt)
print("Recovered plain text:",decrypt_txt)
```

OUTPUT:

```
Enter your plain_txt:mathematics
Cipher Text is : qexliqexmgw
Recovered plain text: mathematics
```

SOURCE CODE:

```
from Crypto.Cipher import DES

def pad(text):

    while len(text)%8!=0:

        text+=b"

    return text

def encrypt(key,plainetxt):

    cipher=DES.new(key,DES.MODE_ECB)

    padded_text=pad(plaintext)

    ciphertext=cipher.encrypt(padded_text)

    return ciphertext

def decrypt(key,ciphertext):

    cipher=DES.new(key,DES.MODE_ECB)

    padded_text=cipher.decrypt(ciphertext)

    plaintext=padded_text.rstrip(b")

    return plaintext
```

```
key=b'secretke'

plaintext=b"This is a secret message"

ciphertext=encrypt(key,plaintext)

decrypt_text=decrypt(key,ciphertext)

print("PlainText:",plaintext)

print("CipherText:",ciphertext)

print("Decrypted Text:",decrypted_text)
```

OUTPUT:

```
Enter the plain text:This is plain text
Enter the key:encrypted
Ciphertext: DDD
```

SOURCE CODE

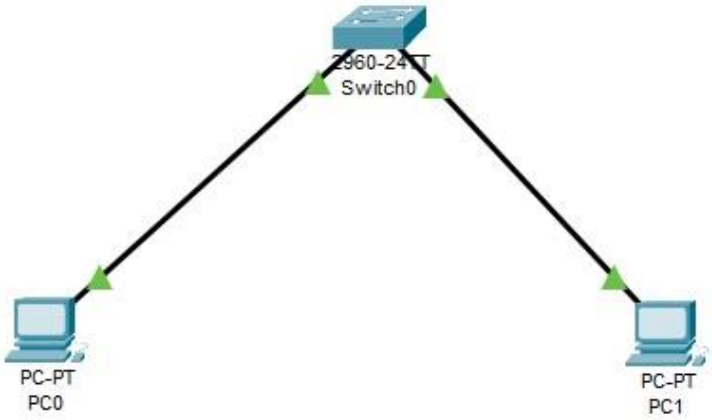
```
import hashlib

def hash_message_sha256(message):

    sha256_hash = hashlib.sha256()

    sha256_hash.update(message.encode('utf-8'))

    hashed_message = sha256_hash.hexdigest()

    return hashed_message

def hash_message_sha1(message):

    sha1_hash = hashlib.sha1()

    sha1_hash.update(message.encode('utf-8')

    hashed_message = sha1_hash.hexdigest()

    return hashed_message

def main():

    message = "This is a sample message to hash"

    sha256_hash = hash_message_sha256(message)
```

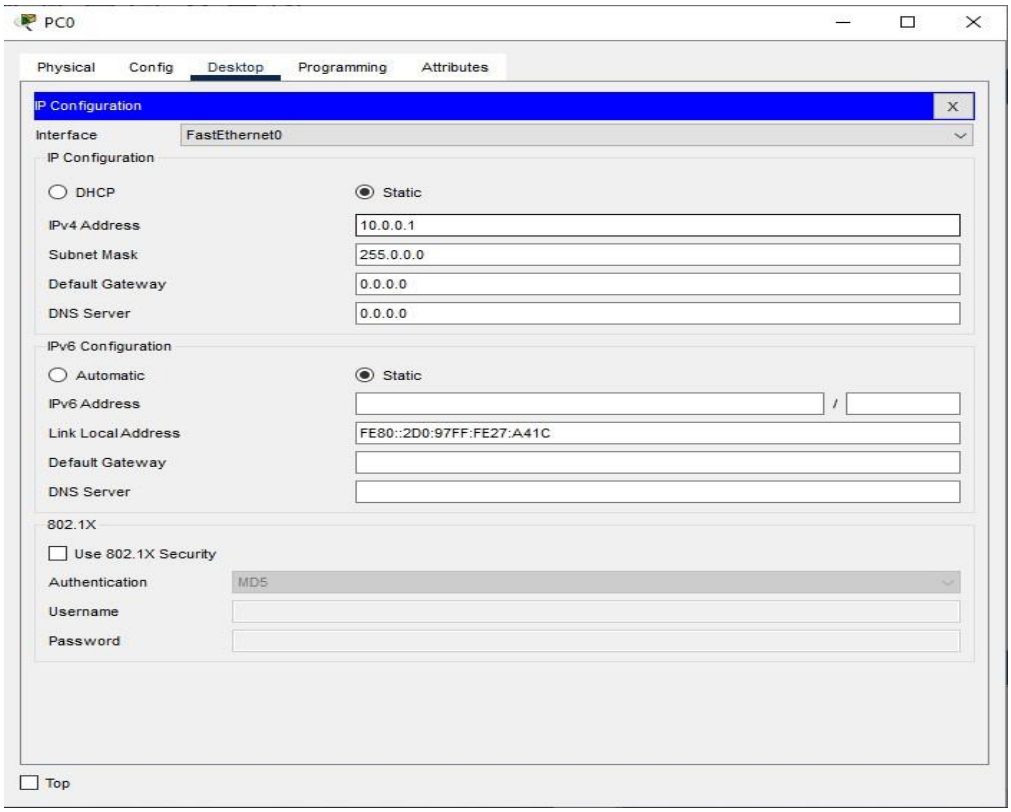
```
    print(f"SHA-256 Hash: {sha256_hash}")

    sha1_hash = hash_message_sha1(message)

    print(f"SHA-1 Hash: {sha1_hash}")

if __name__ == "__main__":

    main()
```

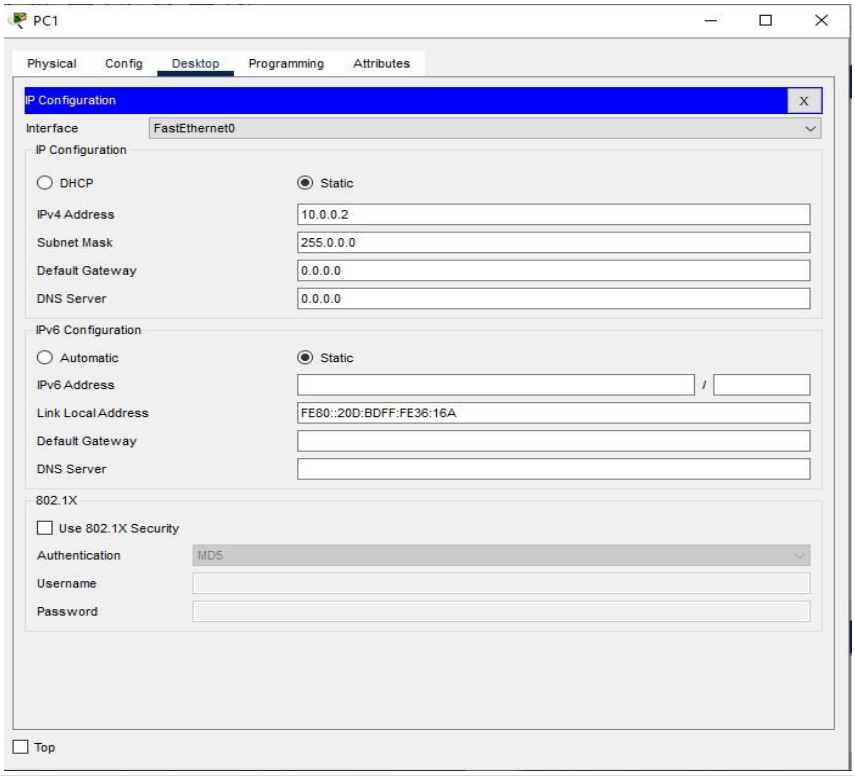
OUTPUT:

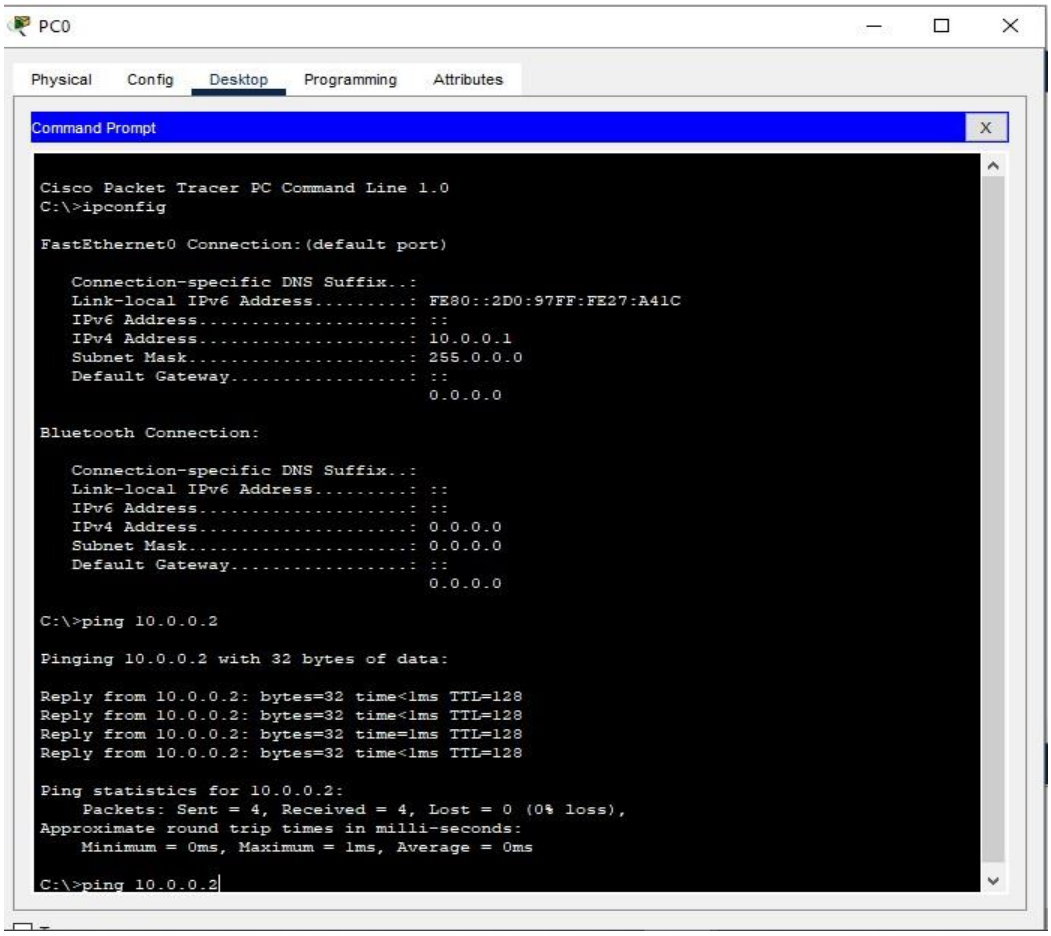
```
SHA-256 Hash: ba5f602c6e26faec77742fdc745f79e14f66ccc3fc420fbaf248bf3e4b77206b
SHA-1 Hash: 1724f8307b0bfbfed07d6e98203d2b82192fd84e
```

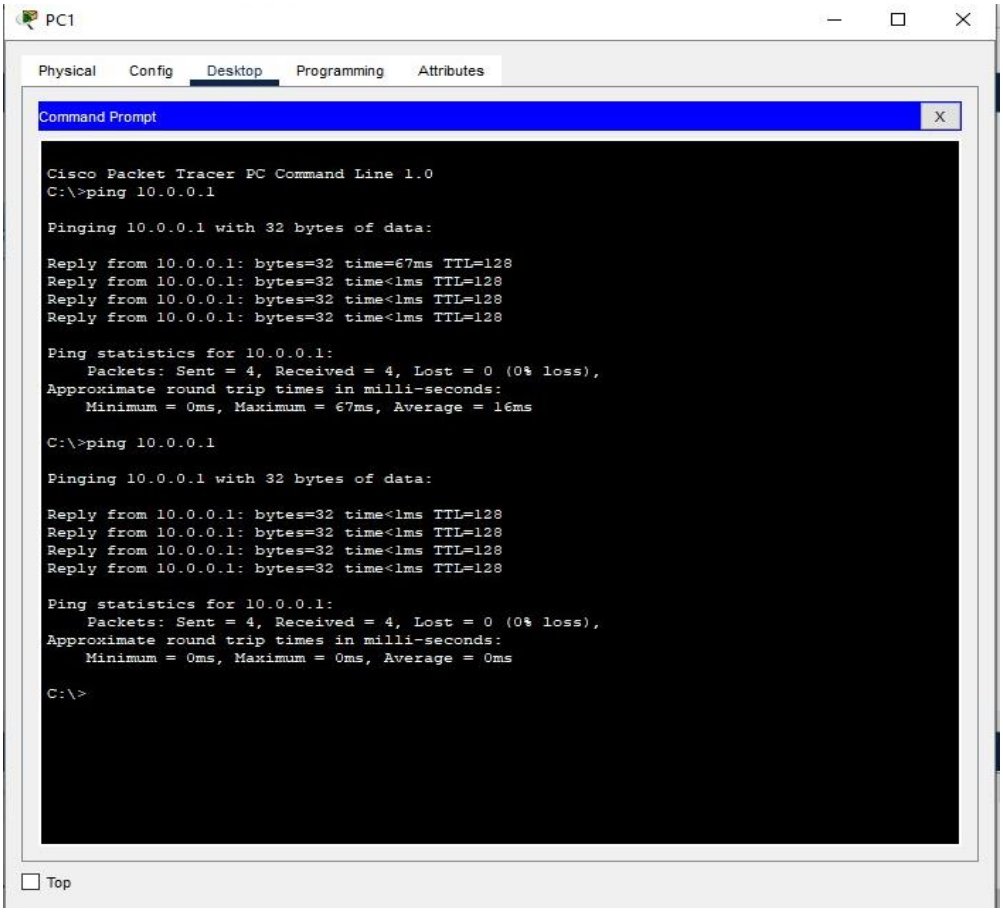
OUTPUT:


2960-24TT
Switch0
PC-PT
PC0
PC-PT
PC1


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
10.0.0.1
Subnet Mask
255.0.0.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2D0:97FF:FE27:A41C
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
10.0.0.2
Subnet Mask
255.0.0.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::20D:BDFF:FE36:16A
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: FE80::2D0:97FF:FE27:A41C
IPv6 Address.....: ::
IPv4 Address.....: 10.0.0.1
Subnet Mask.....: 255.0.0.0
Default Gateway.....: ::
0.0.0.0
Bluetooth Connection:
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0
C:\>ping 10.0.0.2
Pinging 10.0.0.2 with 32 bytes of data:
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time=1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Ping statistics for 10.0.0.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 10.0.0.2


PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.1
Pinging 10.0.0.1 with 32 bytes of data:
Reply from 10.0.0.1: bytes=32 time=67ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Ping statistics for 10.0.0.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 67ms, Average = 16ms
C:\>ping 10.0.0.1
Pinging 10.0.0.1 with 32 bytes of data:
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Ping statistics for 10.0.0.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top

- **Switch#enable**
- **Switch#show ip interface brief**
- **Switch#show port-security**
- **Switch#show port-security address**
- **Switch#show mac-address-table**

```
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0030.a3d0.2216    DYNAMIC     Fa0/10
   1    00d0.9773.de14    DYNAMIC     Fa0/1
```

- **Switch#configure terminal**
- **Switch(config)#interface fastEthernet 0/2**
- **Switch(config-if)#switchport mode access**
- **Switch(config-if)#switchport port-security**
- **Switch(config-if)#switchport port-security mac-address ?**
- **Switch(config-if)#switchport port-security mac-address sticky**
- **Switch(config-if)#switchport port-security maximum ?**
- **Switch(config-if)#switchport port-security maximum 1**
- **Switch(config-if)#switchport port-security violation ?**
- **Switch(config-if)#switchport port-security violation**
- **Switch(config-if)#switchport port-security violation shutdown**
- **Switch(config-if)#^Z**
- **Switch#show port-security**

- **Switch#show port-security address**
- **Go to PC2 command prompt and enter the below thecommand**
    - **ping 10.0.0.1**

```
C:\>ping 10.0.0.1

Pinging 10.0.0.1 with 32 bytes of data:

Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

- **After go to the Switch CLI**
- **Switch#show port-security address**

```
                Secure Mac Address Table
-------------------------------------------------------------------------------
Vlan    Mac Address       Type                          Ports   Remaining Age
                                                                    (mins)
----    -----------       ----                          -----   -------------
   1    000D.BD36.016A    SecureSticky                  Fa0/2        -
-------------------------------------------------------------------------------
```

- **Switch#show port-security**

```
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
                (Count)       (Count)         (Count)
--------------------------------------------------------------------
        Fa0/2        1           1                  0          Shutdown
--------------------------------------------------------------------
```

SOURCE CODE:

```
from cryptography.hazmat.primitives.asymmetric import rsa,padding

from cryptography.hazmat.primitives import hashes



private_key=rsa.generate_private_key(

    public_exponent=65537,

    key_size=2048

    )

public_key=private_key.public_key()



message = b"This is a message to be signed"

hash_algorithm=hashes.SHA256()

hash_object=hashes.Hash(hash_algorithm)

hash_object.update(message)
```

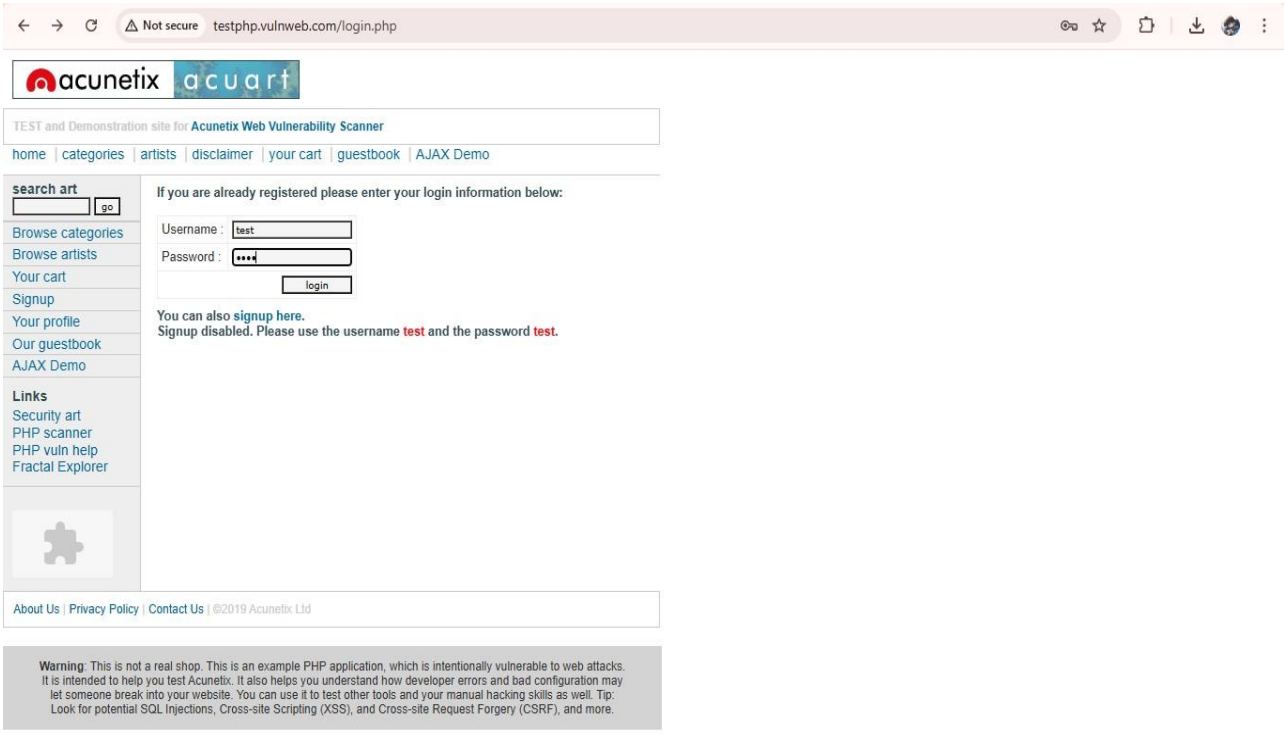
```
digest=hash_object.finalize()


signature=private_key.sign(
    digest,
    padding.PKCS1v15(),
    hashes.SHA256()
)


try:
    public_key.verify(
        signature,
        digest,
        padding.PKCS1v15(),
        hashes.SHA256()
        )
```

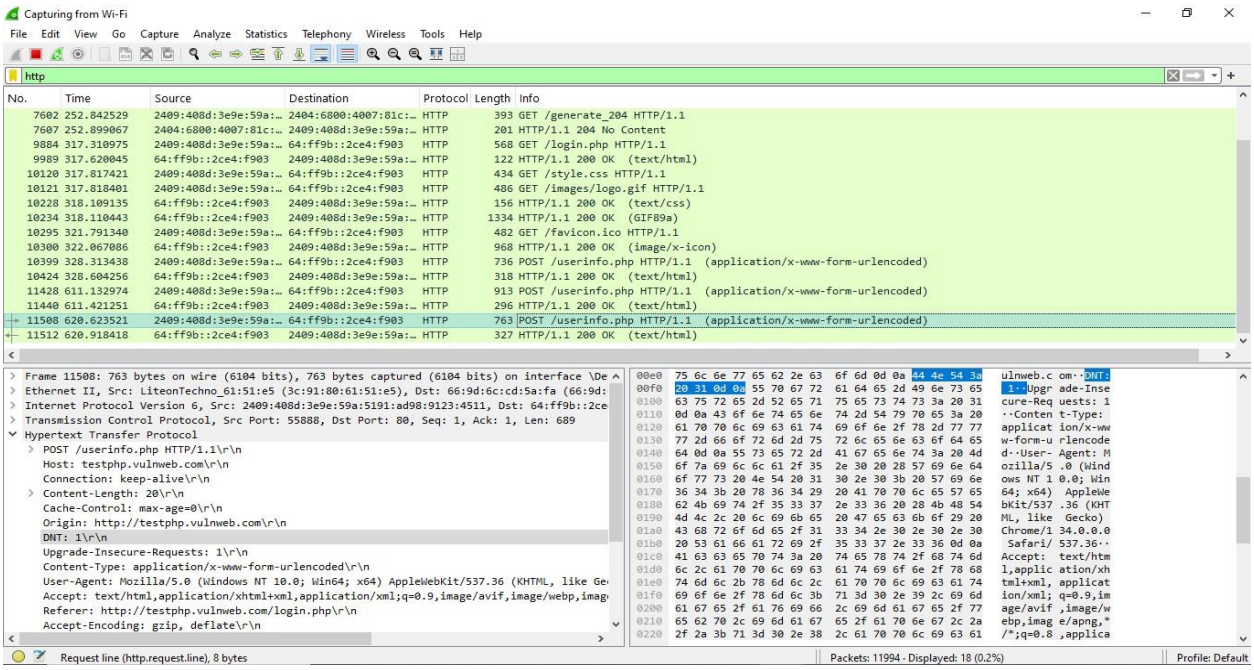
```
    print("The signature is valid.")

except:

    print("The signature is invalid")
```

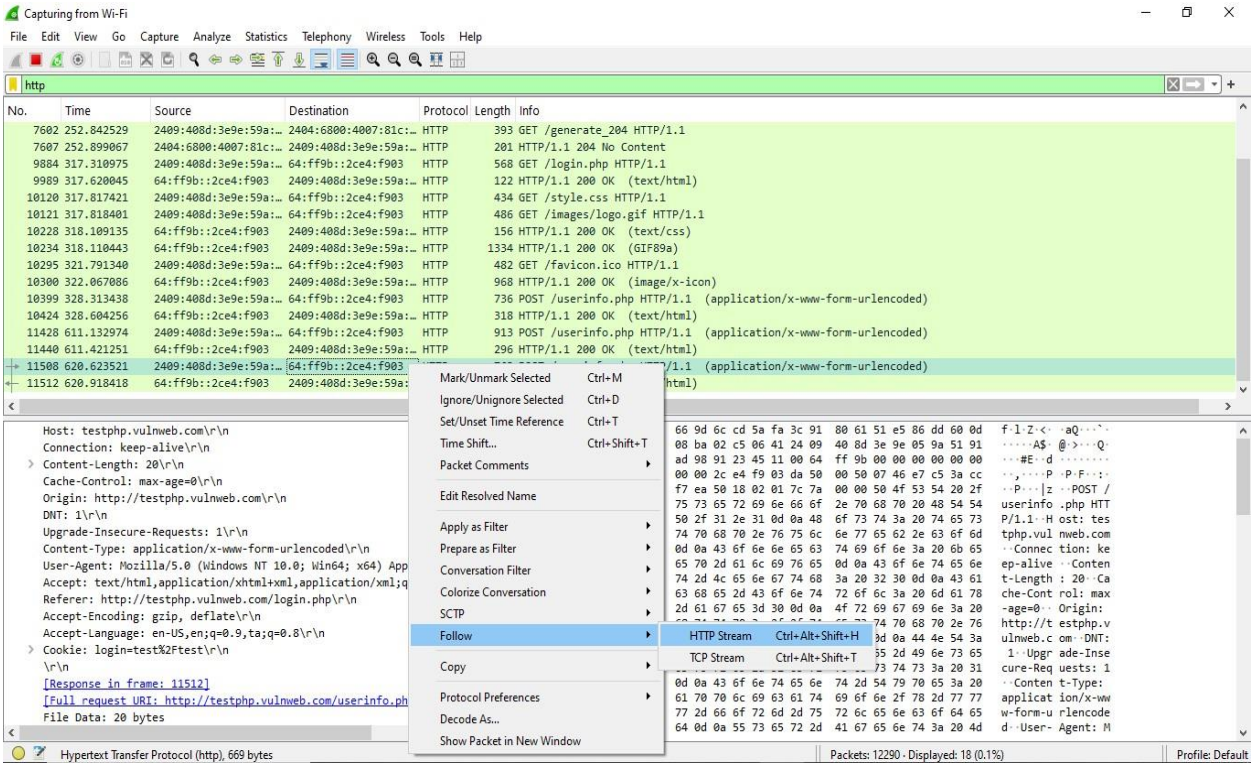
OUTPUT:

```
The signature is valid.
```

OUTPUT:


Not secure testphp.vulnweb.com/login.php
acunetix acuart
TEST and Demonstration site for Acunetix Web Vulnerability Scanner
home | categories | artists | disclaimer | your cart | guestbook | AJAX Demo
search art
go
Browse categories
Browse artists
Your cart
Signup
Your profile
Our guestbook
AJAX Demo
Links
Security art
PHP scanner
PHP vuln help
Fractal Explorer
If you are already registered please enter your login information below:
Username : test
Password :
login
You can also signup here.
Signup disabled. Please use the username test and the password test.
About Us | Privacy Policy | Contact Us | ©2019 Acunetix Ltd
Warning: This is not a real shop. This is an example PHP application, which is intentionally vulnerable to web attacks. It is intended to help you test Acunetix. It also helps you understand how developer errors and bad configuration may let someone break into your website. You can use it to test other tools and your manual hacking skills as well. Tip: Look for potential SQL Injections, Cross-site Scripting (XSS), and Cross-site Request Forgery (CSRF), and more.


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http
No. Time Source Destination Protocol Length Info
7602 252.842529 2409:408d:3e9e:59a:… 2404:6800:4007:81c:… HTTP 393 GET /generate_204 HTTP/1.1
7607 252.899067 2404:6800:4007:81c:… 2409:408d:3e9e:59a:… HTTP 201 HTTP/1.1 204 No Content
9884 317.310975 2409:408d:3e9e:59a:… 64:ff9b::2ce4:f903 HTTP 568 GET /login.php HTTP/1.1
9989 317.620045 64:ff9b::2ce4:f903 2409:408d:3e9e:59a:… HTTP 122 HTTP/1.1 200 OK (text/html)
10120 317.817421 2409:408d:3e9e:59a:… 64:ff9b::2ce4:f903 HTTP 434 GET /style.css HTTP/1.1
10121 317.818401 2409:408d:3e9e:59a:… 64:ff9b::2ce4:f903 HTTP 486 GET /images/logo.gif HTTP/1.1
10228 318.109135 64:ff9b::2ce4:f903 2409:408d:3e9e:59a:… HTTP 156 HTTP/1.1 200 OK (text/css)
10234 318.110443 64:ff9b::2ce4:f903 2409:408d:3e9e:59a:… HTTP 1334 HTTP/1.1 200 OK (GIF89a)
10295 321.791340 2409:408d:3e9e:59a:… 64:ff9b::2ce4:f903 HTTP 482 GET /favicon.ico HTTP/1.1
10300 322.067086 64:ff9b::2ce4:f903 2409:408d:3e9e:59a:… HTTP 968 HTTP/1.1 200 OK (image/x-icon)
10399 328.313438 2409:408d:3e9e:59a:… 64:ff9b::2ce4:f903 HTTP 736 POST /userinfo.php HTTP/1.1 (application/x-www-form-urlencoded)
10424 328.604256 64:ff9b::2ce4:f903 2409:408d:3e9e:59a:… HTTP 318 HTTP/1.1 200 OK (text/html)
11428 611.132974 2409:408d:3e9e:59a:… 64:ff9b::2ce4:f903 HTTP 913 POST /userinfo.php HTTP/1.1 (application/x-www-form-urlencoded)
11440 611.421251 64:ff9b::2ce4:f903 2409:408d:3e9e:59a:… HTTP 296 HTTP/1.1 200 OK (text/html)
11508 620.623521 2409:408d:3e9e:59a:… 64:ff9b::2ce4:f903 HTTP 763 POST /userinfo.php HTTP/1.1 (application/x-www-form-urlencoded)
11512 620.918418 64:ff9b::2ce4:f903 2409:408d:3e9e:59a:… HTTP 327 HTTP/1.1 200 OK (text/html)
Frame 11508: 763 bytes on wire (6104 bits), 763 bytes captured (6104 bits) on interface \De
Ethernet II, Src: LiteonTechno_61:51:e5 (3c:91:80:61:51:e5), Dst: 66:9d:6c:cd:5a:fa (66:9d:
Internet Protocol Version 6, Src: 2409:408d:3e9e:59a:5191:ad98:9123:4511, Dst: 64:ff9b::2ce
Transmission Control Protocol, Src Port: 55888, Dst Port: 80, Seq: 1, Ack: 1, Len: 689
Hypertext Transfer Protocol
POST /userinfo.php HTTP/1.1\r\n
Host: testphp.vulnweb.com\r\n
Connection: keep-alive\r\n
Content-Length: 20\r\n
Cache-Control: max-age=0\r\n
Origin: http://testphp.vulnweb.com\r\n
DNT: 1\r\n
Upgrade-Insecure-Requests: 1\r\n
Content-Type: application/x-www-form-urlencoded\r\n
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Ge
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,imag
Referer: http://testphp.vulnweb.com/login.php\r\n
Accept-Encoding: gzip, deflate\r\n
Request line (http.request.line), 8 bytes
Packets: 11994 · Displayed: 18 (0.2%)
Profile: Default




```
POST /userinfo.php HTTP/1.1
Host: testphp.vulnweb.com
Connection: keep-alive
Content-Length: 20
Cache-Control: max-age=0
Origin: http://testphp.vulnweb.com
DNT: 1
Upgrade-Insecure-Requests: 1
Content-Type: application/x-www-form-urlencoded
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/134.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Referer: http://testphp.vulnweb.com/login.php
Accept-Encoding: gzip, deflate
Accept-Language: en-US,en;q=0.9,ta;q=0.8
Cookie: login=test%2Ftest

uname=test&pass=test
```